

AC-MOT: Optimization-Guided Adaptive Detector Operating-Point Control for Aerial Multi-Object Tracking

Ahmed Gouda Ismail^{1*}, Mohamed S. Mohamed¹ and Tarek Ahmed Mahmoud²

^{1*}Computer Engineering and Artificial Intelligence Department, Military Technical College, Cairo, Egypt.

²Computer Engineering Department, Egypt University of Informatics, Cairo, Egypt.

*Corresponding author(s). E-mail(s): a7medgouda1@gmail.com;
Contributing authors: mohamedms@mtc.edu.eg; tarek.mohamed@eui.edu.eg;

Abstract

Multi-object tracking (MOT) in unmanned aerial vehicle (UAV) videos is challenging due to variations in object density, size, illumination, and blur. However, most tracking-by-detection systems use fixed detector settings that may not suit changing scene conditions. This paper proposes Adaptive Control for Multi-Object Tracking (AC-MOT), a training-free framework that dynamically adjusts the detector operating point, defined by the confidence threshold, non-maximum suppression (NMS) intersection-over-union (IoU) threshold, and input resolution. Using only the current frame and past tracking information, AC-MOT treats the detector operating point as an online control variable without modifying or retraining the detector or tracker. Initial optimization experiments reveal trade-offs among tracking accuracy, identity preservation, and computational efficiency, motivating a three-state controller. On the VisDrone2019 dataset, AC-MOT improves Higher Order Tracking Accuracy (HOTA) from 41.586 to 44.523, Multiple Object Tracking Accuracy (MOTA) from 9.876 to 27.871, and Identity F1 Score (IDF1) from 48.569 to 53.929 with ByteTrack, while reducing identity switches (IDS) from 1,341 to 1,016. With OATrack, MOTA increases from 28.165 to 34.483, and IDS decreases from 773 to 604. Paired bootstrap analysis with 5,000 resamples supports improvements in HOTA, MOTA, IDF1, and IDS for both trackers. Cross-dataset evaluation on the UAVDT dataset, using the same fixed policy without retuning, improves MOTA by 17.601 and 3.267 points with ByteTrack and OATrack, respectively. These results demonstrate that online detector operating-point adaptation can improve aerial MOT performance with limited computational overhead and no additional training.

Keywords: multi-object tracking, UAV video, adaptive inference, detector operating point, real-time image processing

1 Introduction

Unmanned aerial vehicles (UAVs) are widely used for traffic monitoring, surveillance, and search, and many of these tasks need multi-object tracking (MOT): every object must be detected in every frame and keep the same identity over time

[1, 2]. Aerial video makes this hard. Objects are small, their number changes quickly, the camera moves, and illumination and blur change within one video [2–4]. Many of these applications also need results with low delay, so the processing cost matters as much as the accuracy.

The performance of aerial multi-object tracking depends on several factors, including the quality of object detection, the ability to associate objects across consecutive frames, and the computational resources available for processing. These factors are closely related, since errors in object detection can directly affect the tracking process. Missed detections may interrupt object trajectories, while false detections can lead to incorrect associations and identity switches. At the same time, improving detection accuracy often requires additional computation, which may increase processing time and limit real-time performance. Achieving a suitable balance between detection quality, tracking accuracy, identity consistency, and computational efficiency is therefore an important challenge in aerial MOT systems.

Most MOT systems follow the tracking-by-detection design. A detector finds the objects in each frame, and a tracker links the detections over time [5, 6]. Much research improves the tracker, for example its motion model or its association rules, and much research improves the detector itself. Far less attention is given to how the detector is run. In practice the detector uses one confidence threshold, one non-maximum-suppression (NMS) threshold, and one input resolution for the whole video. These three values form the detector operating point, and they decide which boxes reach the tracker.

A fixed operating point does not fit a scene that keeps changing. A setting that works in a sparse, clear scene can keep too much clutter in a dense scene. A setting that works in a dense scene can lose small or weak objects in another part of the same video. A large input helps small objects, but it costs the same time on easy and hard frames. The choice therefore moves false positives, missed objects, identity switches, and runtime together.

Existing adaptive methods only partly address this problem. Adaptive inference methods change the detector’s input scale or configuration, but they usually need an extra learned model and are designed for detection, not tracking [7, 8]. Adaptive tracking methods change thresholds inside one specific tracker [9, 10]. What is missing is a simple, training-free way to adapt the detector operating point online for MOT, without changing the detector or the tracker.

This paper presents AC-MOT, a training-free control layer that adapts the detector operating point online for aerial MOT. It sits between the detector and the tracker, reads inexpensive scene cues, and works with existing detectors and trackers without retraining or modifying them. The method is described in Sect. 3, and Sect. 4 explains how it was developed with optimization and tested step by step.

The main contributions are:

1. A training-free control layer that treats the detector operating point as part of the MOT system and adapts it online (Sect. 3).
2. An optimization-guided development path, in which each step is compared with the baseline it was built on (Sect. 4).
3. An evaluation on VisDrone2019-MOT with two trackers, ByteTrack and the recent OATrack, using eight MOT metrics and paired bootstrap intervals, together with a transfer to UAVDT without retuning (Sect. 5).
4. A latency measurement showing that the control layer adds only a few milliseconds per frame on a Tesla T4 GPU (Sect. 6).

The rest of the paper is organized as follows. Section 2 describes the datasets and evaluation metrics and reviews related work. Section 3 describes AC-MOT, and Section 4 explains how it was developed. Section 5 gives the experimental settings and the results, and Section 6 a discussion. Section 7 concludes the paper.

2 Background and Related Work

This section introduces the datasets and evaluation metrics used in this study, followed by a review of related work on multi-object tracking.

2.1 Dataset Description

Two publicly available aerial video datasets, VisDrone2019-MOT [2] and UAVDT [3], are used to develop and evaluate the proposed AC-MOT framework. Both datasets contain videos captured by unmanned aerial vehicles (UAVs), but they differ in their object categories, scene characteristics, and recording conditions. Using these datasets allows us to examine the performance of AC-MOT under different aerial tracking conditions and to

study whether the same controller can be applied to a new dataset without changing its settings.

2.1.1 VisDrone2019-MOT

VisDrone2019-MOT is a widely used benchmark for multi-object tracking in UAV videos [2]. It contains video sequences recorded by various drone-mounted cameras in 14 cities in China, in urban and rural regions and in both sparse and crowded scenes. The VisDrone benchmark annotates ten object categories, such as pedestrians, cars, buses, and bicycles, with boxes that give the location and identity of each tracked object. Its multi-object tracking track uses 96 video clips: 56 for training (24,198 frames), 7 for validation (2,846 frames), 16 for challenge testing (6,322 frames), and 17 for development testing (6,635 frames), and it evaluates five categories: pedestrian, car, van, bus, and truck [2].

Tracking objects in VisDrone2019-MOT is challenging for several reasons. First, many objects appear small because the camera is positioned at a considerable distance from the ground. Second, the number of visible objects can change significantly between frames, especially in crowded traffic scenes. Third, UAV movement causes changes in camera position and viewing angle, making it difficult to maintain object identities over time. In addition, changes in lighting, background details, and object occlusion can affect detection quality and tracking performance.

These characteristics make VisDrone2019-MOT suitable for studying the effect of detector settings on multi-object tracking. For example, higher input resolution may help detect small objects, while changes in confidence and non-maximum suppression (NMS) thresholds can affect the number of detections passed to the tracker. In this study, VisDrone2019-MOT is used to develop the AC-MOT controller, select its parameters, and evaluate its performance with ByteTrack and OATrack. Separate video sequences are used for controller development and final testing. Specifically, eight training clips serve as calibration data for the final controller (Sect. 4.4), and the seven validation clips serve as evaluation data (Sect. 5).

2.1.2 UAVDT

UAVDT is another benchmark designed for object detection and multi-object tracking in UAV videos, with a particular focus on vehicles [3]. It consists of 100 video sequences selected from more than 10 hours of video recorded at urban locations such as squares, arterial streets, toll stations, highways, crossings, and T-junctions, at 30 frames per second and a resolution of 1080×540 pixels. About 80,000 frames are annotated with 0.84 million boxes of vehicles in three categories: car, truck, and bus. The sequences are recorded under different weather conditions, flying altitudes, and camera viewing angles. These variations produce differences in object size, visibility, background appearance, and the number of vehicles present in each frame. The test set contains 20 sequences for the multi-object tracking task [3], and these 20 sequences are used in this study (Sect. 5).

The two datasets serve different purposes in the experimental evaluation. VisDrone2019-MOT is used for controller development and testing, while UAVDT provides an additional evaluation on a different aerial tracking benchmark. This design helps assess whether the improvements achieved by AC-MOT are limited to the dataset used during development or can also be observed on other aerial videos.

By evaluating AC-MOT with ByteTrack and OATrack on both datasets, the study examines how detector operating-point adaptation affects different tracking algorithms and scene conditions. The results provide evidence of the controller’s effectiveness and its ability to transfer across datasets without requiring changes to the original detector or tracker.

2.2 Multi-Object Tracking Evaluation Metrics

This paper reports HOTA, MOTA, IDF1, IDS, FP, FN, precision, and recall. They are defined as follows [11, 12].

Detection errors. In each frame t , output boxes are matched one to one to ground-truth boxes with an IoU of at least 0.5. Let GT_t be the number of ground-truth boxes, TP_t the number of matched pairs, FP_t the number of output boxes without a match (false positives), and FN_t the number of ground-truth boxes without a

match (false negatives, or missed objects). Over all frames,

$$GT = \sum_t GT_t, \quad TP = \sum_t TP_t, \quad (1)$$

$$FP = \sum_t FP_t, \quad FN = \sum_t FN_t, \quad (2)$$

so that $TP = GT - FN$. Precision is the fraction of output boxes that are correct, and recall is the fraction of ground-truth boxes that are found:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{GT}. \quad (3)$$

Identity switches. Let $\phi_t(i)$ be the output track matched to ground-truth object i in frame t , and $\phi_{\text{prev}}(i)$ the output track to which object i was last matched before t . An identity switch is counted whenever these two tracks differ:

$$IDS = \sum_t \sum_i \mathbf{1}[\phi_t(i) \neq \phi_{\text{prev}}(i)], \quad (4)$$

where the inner sum runs over the objects that are matched in frame t and were matched before.

MOTA and IDF1. MOTA combines the three error types, and IDF1 measures how long each object keeps the correct identity:

$$\text{MOTA} = 1 - \frac{FN + FP + IDS}{GT}, \quad (5)$$

$$\text{IDF1} = \frac{2 \text{IDTP}}{2 \text{IDTP} + \text{IDFP} + \text{IDFN}}. \quad (6)$$

For IDF1, each ground-truth track is first matched to at most one output track over the whole video. IDTP, IDFP, and IDFN are then the true-positive, false-positive, and false-negative boxes under this track-level matching.

HOTA. HOTA balances detection and association [12]. At a localization threshold α , let TP_α , FN_α , and FP_α be the matched pairs, missed ground-truth boxes, and unmatched output boxes. For a matched pair c , $TPA(c)$ is the set of matched pairs with the same ground-truth identity and the same output identity as c ; $FNA(c)$ is the set of boxes of the same ground-truth object that are matched to another output identity or not matched; and $FPA(c)$ is the set of boxes of the same output track that are matched to another

object or not matched. Then

$$\text{DetA}_\alpha = \frac{|TP_\alpha|}{|TP_\alpha| + |FN_\alpha| + |FP_\alpha|}, \quad (7)$$

$$\mathcal{A}(c) = \frac{|TPA(c)|}{|TPA(c)| + |FNA(c)| + |FPA(c)|}, \quad (8)$$

$$\text{AssA}_\alpha = \frac{1}{|TP_\alpha|} \sum_{c \in TP_\alpha} \mathcal{A}(c), \quad (9)$$

$$\text{HOTA} = \frac{1}{19} \sum_\alpha \sqrt{\text{DetA}_\alpha \text{AssA}_\alpha}, \quad (10)$$

where α runs over 0.05, 0.10, ..., 0.95. All scores except the counts IDS, FP, and FN are given in percent.

Paired bootstrap resampling at the sequence level keeps the dependence within each sequence and gives intervals for system differences [13]. Each resample draws the evaluation sequences with replacement. The same draw is used for both systems, the pooled metrics are recomputed, and their difference is recorded. The 95% interval is the 2.5–97.5 percentile range.

2.3 Multi-Object Tracking Literature Survey

Multi-object tracking has been widely studied, with research focusing on improving object detection, data association, and tracking accuracy under different operating conditions. In aerial videos, these tasks become more challenging because of small objects, camera motion, and frequent changes in scene characteristics. Previous studies have addressed these challenges through improved tracking algorithms, adaptive threshold selection, and dynamic inference techniques. In addition, optimization methods and performance evaluation measures have been used to investigate the trade-offs between tracking accuracy and computational efficiency. This section reviews these research directions and discusses their relevance to the problem addressed in this paper.

2.3.1 Tracking-by-Detection Methods in Aerial MOT

Tracking-by-detection is one of the most widely used approaches in multi-object tracking (MOT). In this framework, objects are detected in individual video frames and then associated across

consecutive frames to maintain their identities. Several methods have been developed to improve detection reliability, motion prediction, and data association. However, tracking objects in aerial videos introduces additional challenges that require specialized detection and tracking techniques.

SORT built a compact online pipeline from motion prediction and assignment [5], and StrongSORT adds appearance features to reduce identity fragmentation [14]. ByteTrack first associates high-confidence detections and then uses low-confidence boxes to recover objects that would otherwise be lost [6]. TrackFormer performs detection and association with transformer queries [15]. Standard evaluation follows the MOTChallenge benchmarks [11].

Aerial video adds small targets, changing object counts, camera motion, and clutter [2, 3]. UAVMOT addresses the motion of the UAV camera inside the tracker [16], and LSO-YOLO designs a lightweight real-time detector for small UAV objects [17]. OATrack [18] is a recent UAV small-object tracker published in 2026. It weights association by detection quality and uses a YOLO detection cache. We use OATrack as a second host and keep it unchanged. With the same detector, AC-MOT improves its results in our pipeline: HOTA rises by 1.325, MOTA by 6.318, and IDF1 by 2.258 points, and IDS fall by 169 (Sect. 5). These works improve the tracker or the detector itself. AC-MOT asks a different question: which detector setting should be used for the current scene.

Detector and trackers used in this study.

The experiments use the YOLO11m detector [19], a one-stage detector of the Ultralytics YOLO family, in a public version trained on the VisDrone2019-DET dataset [2]. Two of the trackers above serve as hosts. ByteTrack [6] is a widely used tracker whose second association step uses low-score boxes. OATrack [18] is a recent tracker for small UAV objects that weights association by detection quality. Both hosts are used without modification.

2.3.2 Adaptive Threshold Selection in Multi-Object Tracking

Several studies have investigated adaptive threshold selection to improve multi-object tracking

under changing scene conditions. In conventional tracking-by-detection systems, fixed thresholds are commonly used to determine which detections are accepted and how they are associated with existing tracks. However, the reliability of these thresholds may vary when object density, occlusion, and detection confidence change during a video sequence. Adaptive threshold methods attempt to address this limitation by adjusting selected tracking parameters according to the current conditions.

Adaptive ByteTrack adjusts the confidence threshold used within the ByteTrack [9] algorithm to accommodate variations in detection confidence. This can help the tracker handle objects whose detection scores change because of occlusion, small object size, or changes in image quality. Similarly, AdapTrack [10] adapts the matching thresholds used during data association. By adjusting these thresholds, the tracker can modify its association decisions according to the available detection and tracking information. Although both approaches introduce flexibility into the tracking process, their adaptation mechanisms operate within the tracker and are designed around particular tracking algorithms.

AC-MOT follows a different approach by adapting the detector operating point rather than the internal parameters of the tracker. The detector operating point consists of three parameters: the detection confidence threshold, the non-maximum suppression (NMS) intersection-over-union (IoU) threshold, and the input resolution. Each parameter influences the detections passed to the tracker. The confidence threshold determines which low-confidence detections are retained, the NMS threshold controls the suppression of overlapping bounding boxes, and the input resolution affects the visibility of small objects and the computational cost of detection. Adjusting these parameters together allows the detector configuration to respond to changes in scene conditions before the tracking stage.

An important distinction is that AC-MOT does not require changes to the tracking algorithm. Instead, it operates as an external control mechanism that selects detector configurations using scene information and previous tracking results. The selected configuration is then applied

during detector inference, while the tracker continues to perform its original association and identity management procedures. This design allows AC-MOT to be evaluated with different trackers, including ByteTrack and OATrack, without modifying or retraining either tracker. Therefore, AC-MOT extends adaptive parameter selection beyond tracker-specific thresholds by introducing joint detector operating-point control within the overall MOT pipeline.

2.3.3 Adaptive Inference for Object Detection

Dynamic neural networks change their structure or parameters for each input [20], and resolution-adaptive networks spend less computation on easy inputs [21]. AdaScale selects the input scale of each video frame with a scale regressor trained on detector features [7]. ApproxDet switches detection configurations at run time with accuracy and latency models trained offline [8]. Chameleon re-profiles the configuration of video analytics pipelines over time [22]. These adaptive inference methods mainly aim to improve object detection accuracy, reduce computational cost, or achieve a suitable balance between accuracy and processing time. To select appropriate inference configurations, some approaches rely on additional trained models, such as scale predictors or offline accuracy and latency estimators, while others use periodic profiling to respond to changes in video content. However, their adaptation strategies are generally designed for object detection or video analytics rather than the specific requirements of multi-object tracking, where detection errors can also affect object association and identity consistency. In contrast, AC-MOT focuses on improving tracking performance by dynamically adjusting three detector parameters: the confidence threshold, non-maximum suppression (NMS) IoU threshold, and input resolution. Its controller uses lightweight scene information and previous tracking results to select detector configurations without requiring an additional learned model or retraining the existing detector or tracker. Furthermore, AC-MOT is evaluated using tracking-specific metrics, including tracking accuracy, identity preservation, and association quality, with two different tracking algorithms, ByteTrack and OATrack. Table 1 summarizes

Table 1 Adaptive methods closest to AC-MOT (BT: ByteTrack).

Method	Adapted	Where	Task
AdaScale [7]	input scale	detector	video detection
ApproxDet [8]	detection configuration	detector	video detection
Chameleon [22]	pipeline configuration	analytics pipeline	video analytics
Adapt. BT [9]	confidence threshold	inside BT	MOT
AdapTrack [10]	matching thresholds	inside the tracker	MOT
AC-MOT	confidence, NMS, resolution	detector; tracker unchanged	aerial MOT

the main differences between AC-MOT and the existing adaptive inference approaches.

2.3.4 Optimization and Evaluation for Multi-Object Tracking

Optimization and performance evaluation play an important role in developing reliable multi-object tracking (MOT) systems. The selection of detector and tracking parameters can affect detection accuracy, object association, identity consistency, and computational efficiency. Since improving one aspect of tracking performance may negatively affect another, systematic optimization methods are needed to explore different parameter configurations and identify suitable trade-offs. In addition, evaluating tracking systems requires metrics that measure both detection and identity preservation, together with statistical methods that assess the reliability of observed performance differences.

Optuna provides define-by-run search with the Tree-structured Parzen Estimator (TPE) sampler [23], which was used throughout the development of AC-MOT. The identity-aware study reports a feasible Pareto front based on the usual notion of non-dominance.

3 Proposed AC-MOT Framework

This section presents the proposed AC-MOT framework for adaptive multi-object tracking in aerial videos. The framework dynamically adjusts

the detector operating point according to changing scene conditions without modifying or retraining the detector or tracker. It combines lightweight scene analysis, scene complexity estimation, and a three-state control mechanism to select suitable detector configurations during online tracking. The following subsections describe the operating point, the scene cues, the scene complexity index, the three-state controller, the action map, and how each value of the controller was chosen. Table 2 lists the symbols used in this section.

3.1 Overview and operating point

Figure 1 shows where AC-MOT sits in the pipeline. For each frame, scene analysis and the AC-MOT controller run first and choose the detector settings. The detector and the tracker then run as usual with these settings. The detector settings at frame t are called the operating point

$$\theta_t = (c_t, \eta_t, r_t), \quad (11)$$

which has three parts:

- r_t is the input resolution, the long side of the resized image in pixels. A larger input shows small objects better but takes more time.
- η_t is the NMS IoU threshold. When two boxes overlap with an intersection over union (IoU) above η_t , the box with the lower score is removed. A lower η_t removes more overlapping boxes.
- c_t is the score threshold. Boxes with a detector score below c_t are removed before they reach the tracker. A higher c_t passes fewer but more reliable boxes.

The detector weights and the tracker are frozen; AC-MOT changes only θ_t .

The controller uses only information that is available at run time:

$$\theta_t = \pi(I_t, \mathcal{B}_{1:t-1}), \quad (12)$$

where π is the control rule described in this section, I_t is the image of the current frame, and $\mathcal{B}_{1:t-1}$ are the boxes that the tracker reported for the earlier frames 1 to $t-1$. In words, the decision for frame t uses the current image and the past tracker output. The tracker output of frame t is used only from frame $t+1$ on. No ground truth,

Table 2 Symbols computed at run time. Fixed values are listed in Table 4.

Symbol	Meaning
t	frame index
j	analysis step (frames 1, 11, 21, ...)
I_t	image of frame t
r_t, η_t, c_t	input resolution, NMS IoU threshold, score threshold used at frame t , Eq. (11)
$\mathcal{B}_{1:t-1}$	boxes reported by the tracker for frames 1 to $t-1$
e_t, β_t, ν_t	edge density, brightness, and sharpness of I_t
N_j	number of tracked boxes in frame $t-1$
a_i	area of tracked box i in pixels
C_j	crowd cue, Eq. (13)
T_j	tiny-object ratio, Eq. (14)
S_j	scene complexity index (SCI), Eq. (15)
H_j	the last seven analysis steps
ℓ_j	scene label, Eq. (16)
g_j	target state, Eqs. (17)–(20)
q_j	controller state: LOW, MEDIUM, or HIGH, Eq. (21)
P_j, d_j	peak object count and drop counter, Eqs. (18)–(19)
t_c	frame of the last state change

no future frame, and no detector score enters the decision.

3.2 Scene cues

Scene analysis does not run on every frame. It runs on frame 1 and then once every 10 frames ($t = 1$ or $t \bmod 10 = 1$, that is, frames 1, 11, 21, ...). Each of these frames is an analysis step j . Between two analysis steps the controller keeps its last decision. The stride of 10 frames was selected by a temporal sweep on validation data (Sect. 3.6). Two kinds of cues are computed at each step.

Image cues of the current frame. The frame is reduced to a quarter of its width and height and converted to gray levels, so that the cues are cheap to compute. Three cues are then measured:

- the edge density e_t , the fraction of pixels marked as edges by the Canny detector (lower and upper thresholds 50 and 120); many edges indicate a cluttered scene;
- the brightness β_t , the mean gray level from 0 (black) to 255 (white); a low value indicates a dark or night scene;
- the sharpness ν_t , the variance of the Laplacian of the gray image; a low value indicates a blurred frame.

Box cues of the previous frame. Let N_j be the number of boxes that the tracker reported for

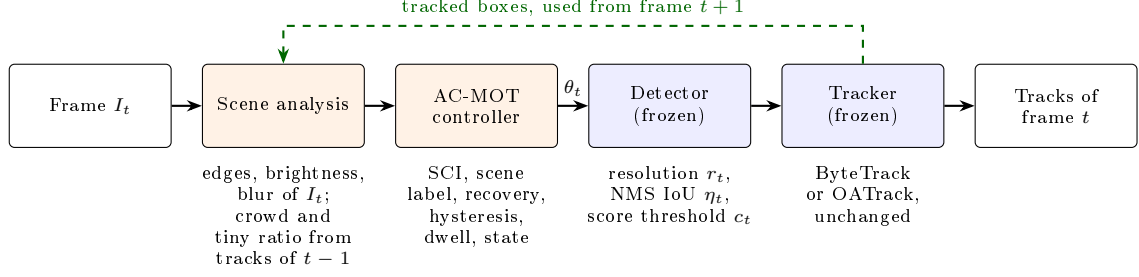


Fig. 1 AC-MOT in a tracking-by-detection pipeline. Scene analysis and the controller choose the operating point $\theta_t = (c_t, \eta_t, r_t)$ before the detector runs. The frozen detector and tracker are not modified. Tracked boxes of frame t feed the scene analysis of later frames (dashed arrow).

frame $t - 1$, and let a_i be the area of box i in pixels of the original image. The crowd cue and the tiny-object ratio are

$$C_j = \min\left(\frac{N_j}{30}, 1\right), \quad (13)$$

$$T_j = \frac{1}{N_j} \sum_{i=1}^{N_j} \mathbf{1}[a_i < 32^2]. \quad (14)$$

C_j is the number of tracked objects divided by 30 and capped at 1, so every frame with 30 or more objects has $C_j = 1$. T_j is the fraction of tracked boxes smaller than 32×32 pixels, which is the size limit of small objects in the COCO benchmark [24]. The indicator $\mathbf{1}[\cdot]$ is 1 when its condition holds and 0 otherwise, and $T_j = 0$ when $N_j = 0$.

3.3 Scene complexity index and scene label

The scene complexity index (SCI) measures how crowded the recent frames are. It is the mean crowd cue of the last seven analysis steps:

$$S_j = \frac{1}{|H_j|} \sum_{k \in H_j} C_k, \quad (15)$$

where H_j holds the steps $j - 6, \dots, j$ (fewer at the start of a video) and $|H_j|$ is their number. With a stride of 10 frames, the seven steps cover about the last 60 frames, so a single unusual frame changes S_j only a little. The window of seven steps was selected by the same temporal sweep as the stride (Sect. 3.6). Because $C_j \leq 1$, S_j lies between 0 and 1. While fewer than 30 objects are tracked,

S_j is simply the recent average number of tracked objects divided by 30.

The SCI uses the crowd cue alone. During development, all 31 combinations of the five cues (crowd, tiny ratio, edges, brightness, and blur) were tested as SCI inputs on the calibration sequences, and the crowd cue alone was selected (Sect. 4.4). The other four cues are still used as checks on the state through the scene label

$$\ell_j = \begin{cases} \text{night} & \text{if } \beta_t < 80, \\ \text{blur} & \text{else if } \nu_t < 180, \\ \text{tiny} & \text{else if } T_j > 0.50, \\ \text{crowded} & \text{else if } C_j > 0.65 \text{ or } e_t > 0.13, \\ \text{clear} & \text{otherwise.} \end{cases} \quad (16)$$

The conditions are checked from top to bottom, and the first one that holds gives the label. For example, $T_j > 0.50$ means that more than half of the tracked boxes are tiny, and $C_j > 0.65$ means that at least 20 objects are tracked.

3.4 Three-state controller

The controller is always in one of three states: LOW, MEDIUM, or HIGH. The names describe how difficult the scene is. They do not rank the input size: in the final action map the HIGH state uses the smallest input (Table 3). The state changes only at analysis steps, in four steps (Algorithm 1).

Step 1: target state. The SCI and the scene label give a target state

$$g_j = \begin{cases} \text{HIGH} & \text{if } S_j > \tau_h \text{ or } T_j > 0.50, \\ \text{MEDIUM} & \text{else if } S_j > \tau_m \text{ or} \\ & \ell_j \in \{\text{crowded}, \text{tiny}\}, \\ \text{LOW} & \text{otherwise,} \end{cases} \quad (17)$$

where $\tau_m = 0.297$ and $\tau_h = 0.511$ are the two SCI thresholds (exact values in Table 4). They were selected by the joint search on the calibration sequences (Sect. 3.6). Since S_j is the recent average number of tracked objects divided by 30, the SCI conditions give MEDIUM above about 9 objects ($30\tau_m = 8.9$) and HIGH above about 15 objects ($30\tau_h = 15.3$).

Step 2: recovery probe. When the number of tracked objects suddenly collapses, the controller tries the HIGH action for a short time. It keeps a slowly decaying peak P_j of the object count and a drop counter d_j :

$$d_j = \begin{cases} d_{j-1} + 1 & \text{if } P_{j-1} \geq 5 \text{ and} \\ & N_j < 0.40 P_{j-1}, \\ 0 & \text{otherwise,} \end{cases} \quad (18)$$

$$P_j = \max(N_j, \lfloor 0.8 P_{j-1} \rfloor), \quad (19)$$

with $P_0 = d_0 = 0$. A collapse is detected when at least 5 objects were seen before and fewer than 40% of them remain. The peak loses 20% at each step, so an old peak is forgotten after a few steps. While $1 \leq d_j \leq 3$, the target g_j is set to HIGH, so the probe lasts at most three analysis steps (30 frames).

Step 3: hysteresis. A move to a lower state is held while the scene is still close to difficult:

$$g_j \leftarrow \begin{cases} \text{HIGH} & \text{if } q_{j-1} = \text{HIGH and} \\ & (S_j > 0.50 \text{ or } T_j > 0.40), \\ \text{MEDIUM} & \text{if } q_{j-1} = \text{MEDIUM,} \\ & g_j = \text{LOW and } S_j > 0.25, \\ g_j & \text{otherwise.} \end{cases} \quad (20)$$

The hold values lie just below the values that lead into each state ($0.50 < \tau_h$, 0.40 below the tiny guard of 0.50, and $0.25 < \tau_m$), so the state does not jump back and forth when the scene stays near a threshold. Upward moves are not held.

Algorithm 1 AC-MOT controller for one video

```

1:  $q \leftarrow \text{LOW}; H \leftarrow \emptyset; P \leftarrow 0; d \leftarrow 0; t_c \leftarrow 1$ 
2: for each frame  $t = 1, 2, \dots$  do
3:   if  $t = 1$  or  $t \bmod 10 = 1$  then
4:      $e_t, \beta_t, \nu_t \leftarrow$  image cues of  $I_t$ 
5:      $N, C, T \leftarrow$  box cues of frame  $t - 1$ 
6:     add  $C$  to  $H$  (keep 7);  $S \leftarrow \text{mean}(H)$ 
7:      $\ell \leftarrow$  label (Eq. 16);  $g \leftarrow$  target (Eq. 17)
8:     update  $d$  and  $P$  (Eqs. 18–19)
9:     if  $1 \leq d \leq 3$  then  $g \leftarrow \text{HIGH}$ 
10:    end if
11:    apply the holds (Eq. 20)
12:    if  $t - t_c \geq 30$  and  $g \neq q$  then
13:       $q \leftarrow g; t_c \leftarrow t$ 
14:    end if
15:  end if
16:   $(r_t, \eta_t, c_t) \leftarrow$  action of state  $q$  (Table 3)
17:  detect with  $(r_t, \eta_t)$ ; drop scores  $< c_t$ 
18:  update the tracker; store its boxes
19: end for

```

Step 4: dwell. After a change, the state is kept for at least 30 frames, that is, three analysis steps:

$$q_j = \begin{cases} g_j & \text{if } t - t_c \geq 30, \\ q_{j-1} & \text{otherwise,} \end{cases} \quad (21)$$

where t_c is the frame of the last state change. The controller starts in LOW with $t_c = 1$, so the first change is possible at frame 31. Whenever $q_j \neq q_{j-1}$, t_c is set to t .

The state q_j then selects the operating point from the action map, and the detector uses it until the next analysis step.

3.5 Action map

Table 3 gives the operating point of each state. LOW uses the largest input (1536 pixels), loose NMS (0.70), and a low score threshold (0.10), so the detector passes more weak boxes when the scene is sparse. MEDIUM uses 1280 pixels, tight NMS (0.45), and a score threshold of 0.40. HIGH uses 1088 pixels, NMS 0.60, and a score threshold of 0.40. In the two difficult states, the detector therefore passes only boxes with a score of at least 0.40. The map was selected together with τ_m and τ_h by the joint search on the calibration sequences (Sect. 3.6). The score threshold c_t is applied on the detector side before the tracker. It is separate

Table 3 Action map. State names describe scene difficulty, not a resolution rank. Selected by the joint search on the calibration sequences.

State	Resolution r	NMS IoU η	Score threshold c
LOW	1536	0.70	0.10
MEDIUM	1280	0.45	0.40
HIGH	1088	0.60	0.40

from any confidence gate inside the tracker, such as the fixed tracker-side gate of OATrack.

3.6 How each value was chosen

All values of Tables 3 and 4 were fixed before the final evaluation and were not changed after it. The last column of Table 4 gives the source of each value. There are four sources.

Joint search on the calibration sequences. The SCI thresholds τ_m and τ_h and the action map were selected together by an Optuna TPE search with 60 trials [23]. The calibration sequences are eight sequences of the VisDrone2019-MOT training split. They serve as the validation (tuning) data of the final stage and are disjoint from the evaluation sequences. Each trial was scored by four-fold cross-validation, with two sequences held out in each fold, as the mean change in HOTA against the baseline. The search ranges were $\tau_m \in [0.10, 0.50]$ and $\tau_h \in [\tau_m + 0.05, \min(\tau_m + 0.40, 0.90)]$, with, for each state, an input of 1088, 1280, or 1536 pixels, an NMS value of 0.45, 0.60, or 0.70, and a score threshold of 0.10, 0.25, or 0.40. The input sizes and NMS values are the operating points for which detector outputs were cached; 1536 pixels with NMS 0.70 is the baseline point, and 0.70 is the default NMS value of the Ultralytics detector. The three score thresholds are the low and high thresholds of ByteTrack (0.10 and 0.25) and the confidence gate of OATrack (0.40). In a separate sweep of the score threshold over 0.01, 0.10, 0.25, and 0.40 on the same sequences, at 1088 pixels and NMS 0.45, 0.40 gave the highest pooled HOTA (+0.881 over 0.01) and improved seven of the eight sequences. The frozen policy reached a mean HOTA gain of 2.447 points over the four folds, with all folds positive (Sect. 4.4).

Cue audit on the calibration sequences. The choice of the crowd cue as the only SCI input came from testing all 31 non-empty subsets of the five cues with the same cross-validation (Sect. 4.4).

Temporal sweep on the validation split. The analysis stride and the SCI window were selected in the first development stage, with the YOLOv8n pipeline, on the VisDrone2019-MOT validation sequences. The sweep tested 25 combinations: windows of 1, 3, 5, 7, and 9 analysis steps and strides of 1, 5, 10, 15, and 20 frames. Five of the 25 settings met both constraints of that stage, at least 25 FPS and at most 271 IDS (the IDS of the original controller on the same data). Among them, a window of 7 and a stride of 10 gave the highest MOTA, 18.165 against 18.044 for the next setting. Operating-point sweeps on the same data covered input resolutions from 512 to 960 pixels, confidence thresholds from 0.05 to 0.50, and NMS thresholds from 0.30 to 0.80, and defined the supported values of the first optimized profiles (Sect. 4).

Fixed design values. The tiny-object area follows the COCO definition of small objects [24]. The remaining values (the crowd normalizer, the Canny thresholds, the label thresholds, the recovery, hold, and dwell values, and the initial state) come from the original AC-MOT controller and were kept unchanged in every search of the final stage. The searched values above were tuned with these values in place, so the selected thresholds and actions are matched to them. For example, the crowd normalizer of 30 only sets the scale of S_j : τ_m and τ_h were searched on this scale and correspond to about 9 and 15 tracked objects.

4 How the Final Controller Was Developed

The controller was developed in four steps: a quality-oriented profile, an identity-aware profile, transfer and cross-pipeline studies, and a modern stage that produced the frozen controller. Each step is compared with the baseline it was built on. The historical steps used other detectors and evaluation protocols than the modern stage, so their numbers are reported in their own tables and are not pooled with the modern results.

4.1 Quality-oriented profile

Baseline. The reference pipeline is the pretrained YOLOv8n detector [25] followed by ByteTrack [6], both at their default settings: confidence threshold 0.25, NMS IoU threshold 0.45, input size

Table 4 Values of the AC-MOT controller and how each value was chosen. Calibration sequences: eight VisDrone2019-MOT training sequences used as tuning data, disjoint from the evaluation sequences. CV: four-fold cross-validation.

Symbol	Meaning	Value	How the value was chosen
τ_m	SCI threshold for MEDIUM, Eq. (17)	0.29747709701135766	joint TPE search on the calibration sequences, 60 trials, CV
τ_h	SCI threshold for HIGH, Eq. (17)	0.5114928923560124	same joint search
(r, η, c)	operating point of each state	Table 3	same joint search over $\{1088, 1280, 1536\} \times \{0.45, 0.60, 0.70\} \times \{0.10, 0.25, 0.40\}$
–	SCI input cue	crowd only	cue audit of all 31 cue subsets on the calibration sequences, CV
–	analysis stride	10 frames	temporal sweep of 25 settings on the validation split; highest MOTA with ≥ 25 FPS and ≤ 271 IDS
$ H_j $	SCI window	7 steps	same temporal sweep
–	tiny-object area	32×32 pixels	COCO small-object limit [24]
–	crowd normalizer, Eq. (13)	30 boxes	original controller; sets the scale on which τ_m, τ_h were searched
–	Canny thresholds (lower, upper)	50, 120	original controller; fixed in all final-stage searches
–	night and blur labels, Eq. (16)	$\beta_t < 80, \nu_t < 180$	original controller; fixed in all final-stage searches
–	crowded label, Eq. (16)	$C_j > 0.65$ or $e_t > 0.13$	original controller; fixed in all final-stage searches
–	tiny label and HIGH guard, Eqs. (16), (17)	$T_j > 0.50$	original controller; fixed in all final-stage searches
–	recovery: minimum peak, drop ratio, peak decay, steps	5, 0.40, 0.80, 3	original controller; fixed in all final-stage searches
–	holds in HIGH (S_j, T_j) and MEDIUM (S_j), Eq. (20)	(0.50, 0.40), 0.25	original controller; fixed in all final-stage searches
–	dwelling, Eq. (21)	30 frames	original controller; fixed in all final-stage searches
–	initial state	LOW	original controller; fixed in all final-stage searches

640 pixels, and the default ByteTrack parameters (high threshold 0.25, low threshold 0.10, new-track threshold 0.25, buffer 30 frames, match threshold 0.80). It runs with one fixed operating point for the whole video.

Method. The first optimization-derived profile kept the same detector, used a tuned ByteTrack profile, and searched only on the VisDrone2019-MOT validation split. Five cues were used: crowd count, the fraction of earlier boxes below 32×32 pixels, edge density, low brightness, and low Laplacian variance. Each cue was normalized with validation statistics, and the index was their weighted sum

$$S_j = \sum_{k \in \{c, t, e, n, b\}} w_k x_{k,j}, \quad (22)$$

where k runs over the five cues (c: crowd, t: tiny ratio, e: edges, n: low brightness, b: blur), $x_{k,j}$ is the normalized value of cue k at analysis step j , and w_k is its weight, with $w_k \geq 0$ and $\sum_k w_k = 1$. The index was averaged over seven analyses taken every ten frames. Confidence and NMS were interpolated linearly between the values at $S = 0$ and $S = 1$, and two sorted thresholds selected one of three resolutions (512, 928, 960 pixels).

An Optuna TPE study [23] jointly searched the five weights, the confidence and NMS end

points, and the two thresholds. The resolution levels, the supported confidence values $\{0.25, \dots, 0.45\}$, the supported NMS values from 0.30 to 0.70, and the temporal settings were fixed beforehand by validation sweeps (Sect. 3.6). The selection rule kept trials with at least 25 FPS and at most 271 IDS (the IDS of the original controller on the same validation data) and then took the highest MOTA, with lower IDS, HOTA, IDF1, and FPS as tie-breaks. The frozen profile (archived as Trial 24) has $w_c = 0.1294928$, $w_t = 0.2217477$, $w_e = 0.4337134$, $w_n = 0.0535577$, $w_b = 0.1614885$, confidence end points (0.30, 0.40), NMS end points (0.35, 0.35), and thresholds (0.1353494, 0.2872868). On validation it reached MOTA 23.038, HOTA 36.110, IDF1 40.758, and IDS 270 at 37.169 FPS.

Comparison. Table 5 compares the profile with the baseline on the historical VisDrone test-dev split, with paired bootstrap intervals (5000 resamples, seed 42). HOTA rises by 5.4051 points, MOTA by 7.2195 points, and IDF1 by 8.8220 points, and throughput rises from 36.528 to 38.985 FPS.

4.2 Identity-aware Pareto profile

Baseline. The baseline is the same YOLOv8n and ByteTrack pipeline at default settings (Sect. 4.1).

Table 5 Quality-oriented profile versus its baseline (YOLOv8n with ByteTrack at default settings) on VisDrone test-dev, historical protocol. Gain: paired bootstrap difference with 95% interval; for IDS it is the reduction.

Metric	Baseline	Quality	Gain [95% CI]
HOTA	28.430	33.835	+5.4051 [4.1273, 6.9022]
MOTA	19.729	26.948	+7.2195 [5.4362, 9.2934]
IDF1	32.724	41.546	+8.8220 [6.9005, 11.0537]
IDS	1235	1184	51 [-114, 219]
FPS	36.528	38.985	-

Method. Quality is not the only deployment goal; identity stability and compute can move in a different direction from MOTA. The second study kept the same cues, operating levels, temporal settings, and validation data, but changed the question. It maximized MOTA and minimized IDS subject only to $\text{FPS} \geq 25$, without the IDS gate. A seeded TPE sampler ran 49 completed trials. A trial dominated another when it had no lower MOTA and no higher IDS, with at least one strict improvement, and the feasible Pareto front was built from these relations. The balanced candidate maximized an equal-weight score of normalized MOTA and normalized IDS reduction, with HOTA, IDF1, FPS, and trial index as tie-breaks.

The selected profile (archived as Trial 22) has weights $w_c = 0.1646453$, $w_t = 0.1746265$, $w_e = 0.5076113$, $w_n = 0.1206956$, $w_b = 0.0324213$, confidence end points (0.40, 0.40), NMS end points (0.60, 0.35), and thresholds (0.2927136, 0.6661672). On validation it reached MOTA 19.330, HOTA 31.651, IDF1 34.226, and IDS 168 at 51.406 FPS.

Comparison. Table 6 compares the profile with the baseline on test-dev. It removes 316 identity switches and raises throughput from 36.528 to 46.024 FPS, while HOTA, MOTA, and IDF1 also rise. Against the quality-oriented profile it has 265 fewer IDS [82, 502] and -2.617 HOTA $[-4.004, -1.694]$. The two profiles therefore answer different questions: one favors overall quality and the other favors identity stability and speed.

Table 6 Identity-aware profile versus its baseline (YOLOv8n with ByteTrack at default settings) on VisDrone test-dev, historical protocol. Gain: paired bootstrap difference with 95% interval; for IDS it is the reduction.

Metric	Baseline	Identity	Gain [95% CI]
HOTA	28.430	31.218	+2.788 [0.747, 4.660]
MOTA	19.729	23.792	+4.063 [0.952, 6.886]
IDF1	32.724	37.870	+5.146 [2.028, 8.080]
IDS	1235	919	316 [175, 470]
FPS	36.528	46.024	-

Table 7 Historical UAVDT transfer without retuning. Baseline: YOLOv8n with ByteTrack at default settings.

Metric	Baseline	Quality	Identity
HOTA	24.085	28.390	26.930
MOTA	13.841	17.399	16.118
IDF1	27.887	34.453	32.014
IDS	558	321	308
FPS	65.015	58.353	61.462

4.3 Transfer and cross-pipeline studies

UAVDT. The baseline is again YOLOv8n with ByteTrack at default settings (Sect. 4.1). Both frozen profiles were applied to UAVDT [3] without retuning (Table 7). The quality-oriented profile raised HOTA by 4.305 points [2.740, 5.709] and the identity-aware profile by 2.845 points [1.238, 4.348]; their IDS reductions are 237 [104, 377] and 250 [110, 404].

U2MOT on VisDrone. The baseline is the published U2MOT tracker [26] with its released YOLOX-X detector, run at its own static setting. Seven VisDrone validation sequences served for development and 17 test-dev sequences for evaluation. The cue weights and transforms of the quality-oriented profile were kept, the tracker was not modified, and the action boundaries were set to the 33rd and 67th percentiles of the index on the U2MOT validation run, without using ground truth. Table 8 gives the comparison: the controlled pipeline has 87 fewer IDS and 1230 fewer FP.

SparseTrack on MOT17. The baseline is the published SparseTrack tracker [27] with a fixed NMS IoU threshold of 0.75, on the seven MOT17 val_half sequences (2652 frames) [11]. A minimal

Table 8 Cross-pipeline study on VisDrone test-dev. Baseline: U2MOT with its YOLOX-X detector at its static setting.

Metric	U2MOT baseline	With controller
MOTA	53.873	53.767
IDF1	69.775	69.849
IDS	1239	1152
FP	41385	40155
FN	63241	64801

Table 9 Cross-pipeline study on MOT17 val_half. Baseline: SparseTrack with a fixed NMS IoU threshold of 0.75.

Metric	SparseTrack baseline	With adaptive NMS
MOTA	76.8881	76.9252
IDF1	81.4943	81.5299
IDS	136	123

edge-driven rule switched only the NMS threshold, between 0.70 and 0.80; its threshold came from a leave-one-sequence-out study on the same sequences. Table 9 gives the comparison: the adaptive rule reduces IDS from 136 to 123, with MOTA +0.0371 points [−0.041, +0.182].

Together these studies show that the preferred profile depends on the objective (quality or identity) and on the tracker host. This motivated a modern stage with a stronger detector, explicit calibration data, and two tracker hosts.

4.4 Modern development of the frozen controller

Baseline. The modern baseline is the YOLO11m detector [19], in the public VisDrone checkpoint described in Sect. 2.3.1, run at a fixed operating point of 1536 pixels and NMS IoU 0.70 with no extra score threshold. It feeds ByteTrack [6] during development and both ByteTrack and OATrack [18] in the final evaluation. A large fixed input is a common choice for small aerial objects when no adaptation is used.

Data and protocol. Eight sequences of the VisDrone2019-MOT training split (3282 frames) were declared as calibration data. They are disjoint from the seven evaluation sequences. All development used four folds, each holding out two calibration sequences, and reported the cross-validated (CV) mean change in HOTA against

Table 10 Main modern development steps on the eight calibration sequences (ByteTrack host). The score-threshold gain is pooled HOTA; the joint-search result is the mean over four folds.

Step	Search	Result
Cue subset	31 subsets	{crowd}
Score threshold	4 values	0.40; +0.881 HOTA
Joint search	60 TPE trials	CV mean +2.447

the baseline. Detector outputs were cached for every resolution and NMS value; a score threshold applied to the cache was checked to reproduce direct inference exactly. Table 10 summarizes the development steps.

Cue audit. Stages 1 and 2 searched the state-to-resolution and state-to-NMS maps on exhaustive grids. Stage 3A searched all 31 non-empty subsets of the five cues and selected {crowd}, which defines the crowd-selected SCI of Eq. (15). The other cues were kept as guard and auxiliary cues of the state logic (Sect. 3).

Score threshold. At 1088 pixels and NMS 0.45, a fixed score threshold of 0.40 raised pooled calibration HOTA by +0.881 over 0.01 and improved seven of the eight sequences. The score threshold was therefore added as the third action dimension.

Final joint search. The final search used 60 TPE trials (seed 43). It searched $\tau_m \in [0.10, 0.50]$, $\tau_h \in [\tau_m + 0.05, \min(\tau_m + 0.40, 0.90)]$, and for each state a resolution in {1088, 1280, 1536}, an NMS value in {0.45, 0.60, 0.70}, and a score threshold in {0.10, 0.25, 0.40}. The objective was the CV mean change in HOTA against the baseline, with ByteTrack as host. Eligible candidates had to realize at least two distinct actions, each on at least 10% of calibration frames, and switch within sequences. The frozen policy is the eligible candidate whose three states use three different actions (Table 3). On calibration data it used LOW, MEDIUM, and HIGH on 7.6%, 10.1%, and 82.3% of frames, with 0.607 state changes per 100 frames on average.

Comparison. Table 11 compares the frozen policy with the baseline on each calibration fold. All four folds are positive, with a CV mean of +2.447 HOTA. After these steps the thresholds, cue subset, and action map were frozen. The final comparison on the evaluation sequences is given in Sect. 5.

Table 11 Frozen policy versus the modern baseline (YOLO11m at 1536 pixels and NMS 0.70, ByteTrack) on the four calibration folds: change in HOTA.

	Fold 1	Fold 2	Fold 3	Fold 4	Mean
Δ HOTA	+0.592	+4.164	+2.029	+3.004	+2.447

5 Results

All experiments use the YOLO11m detector of Sect. 2.3.1. It runs in half precision, and detections of the classes pedestrian, car, van, truck, and bus are passed on. ByteTrack uses the Ultralytics implementation with its default parameters: high threshold 0.25, low threshold 0.10, new-track threshold 0.25, buffer 30 frames, and match threshold 0.80. OATrack is a re-implementation of the published method [18] in its main configuration, with its published tracker-side confidence gate of 0.40. This gate belongs to the tracker and is never changed by AC-MOT. For each host, two systems are compared. The baseline runs the detector at the fixed operating point of 1536 pixels and NMS IoU 0.70, with a minimal score threshold of 0.01. This is the largest input of the operating-point grid, and 0.70 is the default NMS value of the Ultralytics detector. AC-MOT runs the frozen controller of Sect. 3 on the same detector. The two systems differ only in the detector operating point. Uncertainty is estimated with the paired bootstrap of Sect. 2.2, with 5000 resamples (seed 42). Latency is measured on one Tesla T4 GPU over the first 200 frames of sequence `uav0000086_00000_v`, excluding five warm-up frames. For every frame, the time spent in the controller (image statistics, scene analysis, and state update) is recorded separately from the detector and tracker time.

5.1 Frozen VisDrone evaluation

The VisDrone evaluation uses seven sequences of the VisDrone2019-MOT validation split. They contain 71,830 ground-truth boxes. The cue subset, thresholds, and action map of the frozen controller were selected on the separate calibration sequences only (Sect. 4.4). Results are scored with a Python port of the VisDrone2019-MOT Task 4b evaluation. Ignored regions are removed, the five classes above are evaluated separately, CLEAR-MOT and identity measures use IoU

Table 12 Frozen VisDrone evaluation on seven VisDrone2019-MOT sequences (71,830 ground-truth boxes). Baseline: fixed operating point (1536, 0.70).

Metric	ByteTrack		OATrack	
	Baseline	AC-MOT	Baseline	AC-MOT
HOTA	41.586	44.523	46.770	48.095
MOTA	9.876	27.871	28.165	34.483
IDF1	48.569	53.929	57.107	59.364
IDS	1341	1016	773	604
FP	39274	24892	25394	21400
FN	24121	25902	25432	25057
Precision	54.849	64.852	64.628	68.609
Recall	66.419	63.940	64.594	65.116

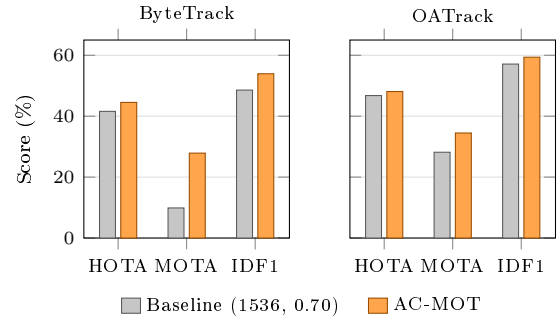


Fig. 2 HOTA, MOTA, and IDF1 on the seven VisDrone sequences for both tracker hosts (values in Table 12).

0.5, and counts are summed over classes and sequences. HOTA is added with the same class-aware scheme using TrackEval. The bootstrap draws seven sequences with replacement.

Table 12 lists the aggregate results on the seven VisDrone sequences, and Fig. 2 shows HOTA, MOTA, and IDF1.

ByteTrack host. AC-MOT changes HOTA from 41.586 to 44.523, MOTA from 9.876 to 27.871, and IDF1 from 48.569 to 53.929. IDS fall from 1341 to 1016 and FP from 39274 to 24892, while FN rise from 24121 to 25902. Precision rises from 54.849 to 64.852 and recall falls from 66.419 to 63.940. The detector output becomes more selective: clutter and identity-confusing boxes are removed at a small cost in recall.

OATrack host. AC-MOT changes HOTA from 46.770 to 48.095, MOTA from 28.165 to 34.483, and IDF1 from 57.107 to 59.364. IDS fall from 773 to 604, FP from 25394 to 21400, and FN from 25432 to 25057. Precision rises from 64.628 to 68.609 and recall from 64.594 to 65.116. OATrack starts from a stronger baseline.

Table 13 Paired sequence-level bootstrap on VisDrone: AC-MOT minus baseline, 5000 resamples, seed 42. HOTA, MOTA, IDF1, precision, and recall in points; IDS, FP, and FN in counts.

Host	Metric	Difference	95% interval
ByteTrack	HOTA	+2.937	[+2.356, +3.375]
	MOTA	+17.995	[+13.541, +23.282]
	IDF1	+5.360	[+4.490, +6.178]
	IDS	-325	[-498, -164]
	FP	-14382	[-19176, -9620]
	FN	+1781	[+288, +3594.075]
	Precision	+10.003	[+8.712, +11.895]
	Recall	-2.479	[-4.507, -0.469]
OATrack	HOTA	+1.325	[+0.200, +2.264]
	MOTA	+6.318	[+4.741, +7.879]
	IDF1	+2.258	[+0.548, +3.474]
	IDS	-169	[-292, -42]
	FP	-3994	[-6190, -2494.900]
	FN	-375	[-1462.575, +670]
	Precision	+3.981	[+2.957, +5.511]
	Recall	+0.522	[-0.869, +2.111]

Paired bootstrap. Table 13 gives the paired differences (AC-MOT minus baseline) and their 95% intervals. With ByteTrack, the intervals of HOTA, MOTA, IDF1, precision, IDS, and FP lie entirely on the side of improvement, and FN and recall move in the opposite direction. With OATrack, HOTA, MOTA, IDF1, and precision rise and IDS and FP fall, all with intervals on one side of zero.

Controller behaviour. The controller uses all three states with both hosts. With ByteTrack, LOW, MEDIUM, and HIGH cover 7.4%, 30.1%, and 62.5% of the frames; with OATrack they cover 7.4%, 33.3%, and 59.3%. The two hosts diverge through their different tracked boxes. State changes are rare, between 0.10 and 0.54 per 100 frames depending on the sequence, so the detector input stays stable over long stretches of video.

5.2 Frozen UAVDT transfer

The UAVDT transfer uses 20 test sequences with 16,592 frames, scored with one vehicle class. The frozen controller, thresholds, and action map are applied without any change. Detector outputs for UAVDT were produced at the three frozen operating points, giving 49,776 detector forward passes. UAVDT is a single frozen run per host. The paired bootstrap draws 20 sequences with replacement from these runs. It is computed from the stored

Table 14 Frozen UAVDT transfer without retuning (20 sequences, 16,592 frames, one vehicle class). Baseline: fixed operating point (1536, 0.70).

Metric	ByteTrack		OATrack	
	Baseline	AC-MOT	Baseline	AC-MOT
HOTA	45.923	47.004	47.728	47.640
MOTA	-1.573	16.028	19.607	22.873
IDF1	58.400	61.767	62.101	62.931
IDS	601	486	517	486
FP	279616	212077	197937	183669
FN	66050	73702	75612	78774
Precision	49.571	55.751	57.270	58.800
Recall	80.625	78.381	77.820	76.893

Table 15 Paired sequence-level bootstrap on UAVDT: AC-MOT minus baseline, 5000 resamples, seed 42, 20 sequences. MOTA and IDF1 in points; IDS, FP, and FN in counts.

Host	Metric	Difference	95% interval
ByteTrack	MOTA	+17.601	[+11.134, +26.767]
	IDF1	+3.367	[+1.676, +5.013]
	IDS	-115	[-211.025, -26.975]
	FP	-67539	[-104215.350, -37352.950]
	FN	+7652	[+1542.925, +18359.400]
OATrack	MOTA	+3.267	[+0.607, +5.881]
	IDF1	+0.831	[+0.125, +1.271]
	IDS	-31	[-101, +39]
	FP	-14268	[-33958.075, -2188.950]
	FN	+3162	[-211.100, +8959.050]

per-sequence counts of the two systems and therefore covers the count-based metrics MOTA, IDF1, IDS, FP, and FN.

Table 14 and Fig. 3 report the transfer to UAVDT without retuning. With ByteTrack, HOTA changes from 45.923 to 47.004 (+1.081), MOTA from -1.573 to 16.028 (+17.601), and IDF1 from 58.400 to 61.767 (+3.367). IDS fall by 115 and FP by 67539, while FN rise by 7652. With OATrack, MOTA changes from 19.607 to 22.873 (+3.267), IDF1 from 62.101 to 62.931 (+0.831), IDS fall by 31, and FP fall by 14268. With both hosts the transfer gives fewer FP and IDS and higher precision, with slightly lower recall, the same pattern as with ByteTrack on VisDrone.

Table 15 gives the paired bootstrap differences on UAVDT. With ByteTrack, the intervals of MOTA, IDF1, IDS, and FP lie entirely on the side of improvement, and FN moves in the opposite direction. With OATrack, MOTA and IDF1 rise and FP fall, with intervals on one side of zero, while the intervals of IDS and FN include zero.

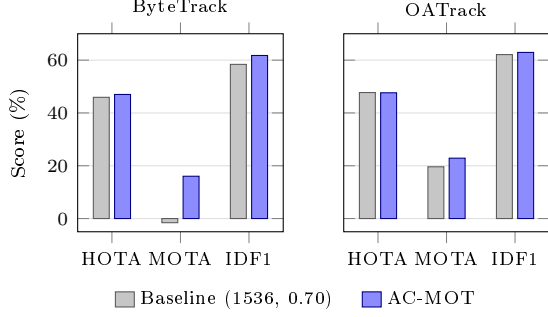


Fig. 3 HOTA, MOTA, and IDF1 of the frozen UAVDT transfer for both tracker hosts (values in Table 14).

6 Discussion

The operating point is part of the MOT system. The same frozen detector and the same trackers give very different MOT results at different operating points. Moving away from the fixed 1536/0.70 point raises MOTA by 17.995 points with ByteTrack and by 6.318 points with OATrack on VisDrone. The detector operating point is therefore a first-order variable of aerial MOT and deserves the same attention as the association method.

Selective observation streams. The main effect of AC-MOT is a more selective detection stream. FP and IDS drop sharply, precision rises, and FN rise a little with ByteTrack. MOTA reacts most strongly because it counts FP directly; HOTA and IDF1 rise by smaller amounts. The same pattern appears on UAVDT, where FP fall by 67539 with ByteTrack and by 14268 with OATrack.

Tracker dependence. ByteTrack gains more than OATrack. ByteTrack starts from a weaker baseline and uses low-score boxes in its second association step, so a likely reason is that removing clutter helps it directly. OATrack already weights detections by quality inside its association, and it gains mainly in MOTA, IDF1, IDS, and FP. Results are therefore reported per host.

Development path. The historical profiles showed that the preferred operating point depends on the goal: quality or identity stability and speed. The modern stage kept a finite, readable controller with one crowd index, a few guard cues, three states, and three actions.

Deployment. AC-MOT needs no training and no access to detector or tracker internals. It changes three arguments of the detector call. Its

analysis runs on a quarter-scale image every ten frames and uses boxes that the tracker already produced. On a Tesla T4 the controller takes 4.26 ms per frame with ByteTrack and 4.20 ms with OATrack, which is small next to detector inference. The control layer therefore adds very little latency and is suitable for real-time use. The same interface can host other action maps, for example maps chosen for a latency budget, without changes to the detector or the tracker.

7 Conclusion

This paper presented Adaptive Control for Multi-Object Tracking (AC-MOT), a training-free method that automatically adjusts detector settings to improve multi-object tracking in videos captured by unmanned aerial vehicles (UAVs). Unlike traditional tracking systems that use fixed detector settings, AC-MOT changes the detection confidence threshold, non-maximum suppression (NMS) IoU threshold, and input resolution based on changes in the scene. The method uses only the current video frame and previous tracking results, without changing or retraining the detector or tracker. It uses a scene complexity index based on object density, together with other scene features such as small objects, illumination, blur, and edges. A three-state controller, supported by recovery, hysteresis, and dwell-time rules, helps the system adjust its settings smoothly and avoid frequent changes.

The controller was developed through several optimization experiments to study how different detector settings affect tracking accuracy, object identity, and processing time. The results showed that the best detector settings depend on the tracking method and the performance goal. The final controller settings were selected using sequence-level cross-validation on separate calibration videos and were then kept fixed during testing.

Experiments on seven VisDrone2019-MOT video sequences showed that AC-MOT improved Higher Order Tracking Accuracy (HOTA) by 2.937 points with ByteTrack and 1.325 points with OATrack. Multiple Object Tracking Accuracy (MOTA) also increased by 17.995 and 6.318 points, respectively. Paired bootstrap analysis was used to calculate confidence intervals for the performance differences. The same controller was also

tested on the UAVDT dataset without changing its settings, improving MOTA by 17.601 points with ByteTrack and 3.267 points with OATrack.

Overall, the results show that adjusting detector settings during video processing can improve aerial multi-object tracking without changing the detector or tracker. They also show that detector settings should be considered an important part of the tracking system rather than fixed values used for all video frames. Future work will focus on improving the controller and testing it with more detectors, tracking methods, and aerial video datasets.

Declarations

Funding. This research received no external funding.

Conflict of interest. The authors declare no competing interests.

Ethics approval and consent to participate. This computational study uses public benchmark data and does not involve human participants or animals.

Consent for publication. Not applicable.

Data and code availability. The frozen controller specification, result records, evidence map, staged sections, scripts, and build package are included in the project repository. Dataset redistribution remains subject to the original benchmark licenses and access terms.

Author contributions. Ahmed Gouda Ismail: Conceptualization, Methodology, Software, Investigation, Data curation, Formal analysis, Visualization, Writing – original draft, Writing – review & editing. Mohamed S. Mohamed: Supervision, Methodology, Writing – review & editing. Tarek Ahmed Mahmoud: Supervision, Validation, Writing – review & editing.

Acknowledgements. Not applicable.

Submission exclusivity. The manuscript is not under consideration elsewhere and has not been published previously.

AI use in manuscript preparation. Generative AI tools were used to assist with drafting and language editing. The authors reviewed and verified all content, numerical values, and figures and take full responsibility for the manuscript.

References

- [1] Wu, X., Li, W., Hong, D., Tao, R., Du, Q.: Deep Learning for Unmanned Aerial Vehicle-Based Object Detection and Tracking: A survey. *IEEE Geoscience and Remote Sensing Magazine* **10**(1), 91–124 (2022) <https://doi.org/10.1109/mgrs.2021.3115137>
- [2] Zhu, P., Wen, L., Du, D., Bian, X., Fan, H., Hu, Q., Ling, H.: Detection and Tracking Meet Drones Challenge. *IEEE Transactions on Pattern Analysis and Machine Intelligence* **44**(11), 7380–7399 (2022) <https://doi.org/10.1109/tpami.2021.3119563>
- [3] Du, D., Qi, Y., Yu, H., Yang, Y., Duan, K., Li, G., Zhang, W., Huang, Q., Tian, Q.: The Unmanned Aerial Vehicle Benchmark: Object Detection and Tracking. In: *Computer Vision – ECCV 2018. Lecture Notes in Computer Science*, pp. 375–391. Springer, Cham (2018). https://doi.org/10.1007/978-3-030-01249-6_23
- [4] Cheng, G., Yuan, X., Yao, X., Yan, K., Zeng, Q., Xie, X., Han, J.: Towards Large-Scale Small Object Detection: Survey and Benchmarks. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 1–20 (2023) <https://doi.org/10.1109/tpami.2023.3290594>
- [5] Bewley, A., Ge, Z., Ott, L., Ramos, F., Upcroft, B.: Simple online and realtime tracking. In: *2016 IEEE International Conference on Image Processing (ICIP)*, pp. 3464–3468. IEEE, Phoenix, AZ, USA (2016). <https://doi.org/10.1109/icip.2016.7533003>
- [6] Zhang, Y., Sun, P., Jiang, Y., Yu, D., Weng, F., Yuan, Z., Luo, P., Liu, W., Wang, X.: ByteTrack: Multi-object Tracking by Associating Every Detection Box. In: *Computer Vision – ECCV 2022. Lecture Notes in Computer Science*, pp. 1–21. Springer, Cham (2022). https://doi.org/10.1007/978-3-031-20047-2_1
- [7] Chin, T.-W., Ding, R., Marculescu, D.: AdaScale: Towards Real-time Video Object

- Detection using Adaptive Scaling. In: Proceedings of Machine Learning and Systems, vol. 1, pp. 431–441 (2019)
- [8] Xu, R., Zhang, C.-l., Wang, P., Lee, J., Mitra, S., Chatterji, S., Li, Y., Bagchi, S.: ApproxDet: Content and Contention-Aware Approximate Object Detection for Mobiles. In: Proceedings of the 18th Conference on Embedded Networked Sensor Systems (SenSys), pp. 449–462. ACM (2020). <https://doi.org/10.1145/3384419.3431159>
- [9] Ma, L.V., Hussain, M.I., Park, J., Kim, J., Jeon, M.: Adaptive Confidence Threshold for ByteTrack in Multi-Object Tracking. In: 2023 12th International Conference on Control, Automation and Information Sciences (ICCAIS), pp. 370–374. IEEE (2023). <https://doi.org/10.1109/iccais59597.2023.10382403>
- [10] Shim, K., Ko, K., Hwang, J., Kim, C.: AdapTrack: Adaptive Thresholding-Based Matching for Multi-Object Tracking. In: 2024 IEEE International Conference on Image Processing (ICIP), pp. 2222–2228. IEEE (2024). <https://doi.org/10.1109/icip51287.2024.10648139>
- [11] Dendorfer, P., Ošep, A., Milan, A., Schindler, K., Cremers, D., Reid, I., Roth, S., Leal-Taixé, L.: MOTChallenge: A Benchmark for Single-Camera Multiple Target Tracking. International Journal of Computer Vision **129**(4), 845–881 (2021) <https://doi.org/10.1007/s11263-020-01393-0>
- [12] Luiten, J., Ošep, A., Dendorfer, P., Torr, P., Geiger, A., Leal-Taixé, L., Leibe, B.: HOTA: A Higher Order Metric for Evaluating Multi-object Tracking. International Journal of Computer Vision **129**(2), 548–578 (2021) <https://doi.org/10.1007/s11263-020-01375-2>
- [13] Efron, B., Tibshirani, R.J.: An Introduction to the Bootstrap. Springer, New York (1993). <https://doi.org/10.1007/978-1-4899-4541-9>
- [14] Du, Y., Zhao, Z., Song, Y., Zhao, Y., Su, F., Gong, T., Meng, H.: StrongSORT: Make DeepSORT Great Again. IEEE Transactions on Multimedia **25**, 8725–8737 (2023) <https://doi.org/10.1109/tmm.2023.3240881>
- [15] Meinhardt, T., Kirillov, A., Leal-Taixé, L., Feichtenhofer, C.: TrackFormer: Multi-Object Tracking With Transformers. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 8844–8854 (2022)
- [16] Liu, S., Li, X., Lu, H., He, Y.: Multi-Object Tracking Meets Moving UAV. In: 2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 8866–8875. IEEE (2022). <https://doi.org/10.1109/cvpr52688.2022.00867>
- [17] Dai, P., Chen, L., Lian, Y., Xie, L.: LSO-YOLO: a lightweight real-time object detection network for UAV small-object detection. Journal of Real-Time Image Processing **23**(5), 176 (2026) <https://doi.org/10.1007/s11554-026-01977-y>
- [18] Qi, L., Wang, L., Jiang, S.: OATrack: A Quality-Gated Progressive Association Framework with a YOLO Detection Cache for UAV Small-Object Multi-Object Tracking. Sensors **26**(16), 5222 (2026) <https://doi.org/10.3390/s26165222>
- [19] Jocher, G., Qiu, J.: Ultralytics YOLO11. <https://github.com/ultralytics/ultralytics>. Version 11.0.0 (2024)
- [20] Han, Y., Huang, G., Song, S., Yang, L., Wang, H., Wang, Y.: Dynamic Neural Networks: A Survey. IEEE Transactions on Pattern Analysis and Machine Intelligence **44**(11), 7436–7456 (2022) <https://doi.org/10.1109/tpami.2021.3117837>
- [21] Yang, L., Han, Y., Chen, X., Song, S., Dai, J., Huang, G.: Resolution Adaptive Networks for Efficient Inference. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 2369–2378 (2020)
- [22] Jiang, J., Ananthanarayanan, G., Bodik, P.,

- Sen, S., Stoica, I.: Chameleon: Scalable Adaptation of Video Analytics. In: Proceedings of the 2018 Conference of the ACM Special Interest Group on Data Communication, pp. 253–266. ACM, New York, NY, USA (2018). <https://doi.org/10.1145/3230543.3230574>
- [23] Akiba, T., Sano, S., Yanase, T., Ohta, T., Koyama, M.: Optuna: A Next-generation Hyperparameter Optimization Framework. In: Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp. 2623–2631. ACM, Anchorage, AK, USA (2019). <https://doi.org/10.1145/3292500.3330701>
- [24] Lin, T.-Y., Maire, M., Belongie, S., Hays, J., Perona, P., Ramanan, D., Dollár, P., Zitnick, C.L.: Microsoft COCO: Common Objects in Context. In: Computer Vision – ECCV 2014. Lecture Notes in Computer Science, pp. 740–755. Springer, Cham (2014). https://doi.org/10.1007/978-3-319-10602-1_48
- [25] Jocher, G., Chaurasia, A., Qiu, J.: Ultralytics YOLOv8. <https://github.com/ultralytics/ultralytics>. Version 8.0.0 (2023)
- [26] Liu, K., Jin, S., Fu, Z., Chen, Z., Jiang, R., Ye, J.: Uncertainty-aware Unsupervised Multi-Object Tracking. In: Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), pp. 9996–10005 (2023)
- [27] Liu, Z., Wang, X., Wang, C., Liu, W., Bai, X.: SparseTrack: Multi-Object Tracking by Performing Scene Decomposition Based on Pseudo-Depth. *IEEE Transactions on Circuits and Systems for Video Technology* **35**(5), 4870–4882 (2025) <https://doi.org/10.1109/TCSVT.2024.3524670>